

Team SHIN: fEMR Sprint Review #1



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Product Vision / Backlog

- We envision a functional CSV import feature that can bulk load formulary data into the fEMR kit, updating existing medication data and adding new rows where relevant
- The formulary should also include a burn rate feature that illustrates usage rate and provides a shopping list for needed medications
- As a stretch goal, we hope to implement fuzzy searching into the search feature (deal with potential misspelling of medication)

Sprint #1 Goal

- Goal 1: Set up our environments and ensure we can navigate the software

Challenges : Some of us had some issues with installing the Scala plugin.

- Goal 2: Analyze the current CSV implementation, make note of any pain points/places for improvement, and determine if we wanted to build atop the code, heavily modify it or rebuild from scratch

Technology Decisions

- For MacOS Silicone computers, Play Framework Plugin was not showing up in IntelliJ Marketplace Plugin search -> JetBrains discontinued the separate Play Framework plugin for IntelliJ IDEA 2026.2.3 (current version)
 - It is only compatible through *IntelliJ IDEA 2023.1*
<https://www.jetbrains.com/idea/download/other/#releases-2023>
- Potential library options for parsing CSV
 - Apache Commons CSV
 - Jackson library for converting CSV to JSON

Reflecting

- Issues with the current CSV implementation
 - No obvious format/schema given for the uploader to follow. A key example of what headers in the CSV should be would be helpful to the user
 - To the extent of our current testing, the uploading a CSV results in an error. Given this, we intend to rebuild, or at least heavily modify the original code to make it functional.
 - The UI of the CSV branch is 5 years out of date. We hope to integrate CSV logic with the new cleaner UI
- Use all Jira features -> put all tasks on backlog, set priorities, log hours

Next Sprint Tasks

Tasks assuming we rebuild from scratch

- Note: If we can get the original implementation working, some of these tasks may prove unnecessary
- We will dedicate an 1-2 more hours of time trying to make the original code functional, before rebuilding
- 1. Add a button that accepts CSV file input 2
- 2. Parse the CSV input into a format suited for reading in Java and uploading to the database 5
- 3. Add error checking and validation to ensure data is formatted properly and can translated correctly to the database schema 4
- 4. Input the parsed data into the database 3
- 5. Create destination screens/popup based on the results 3
 - a. Upload success screen
 - b. Upload error screen
 - c. Invalid format screen
 - d. Invalid row(s) screen
- 6. Highlight updated cells and cells with issues

Future Sprint Tasks

1. Long-term ideas:
 - a. Error-checking for misspelled medication
2. Look into implementing burn-down rate

Design Decisions

- 1) Only the existing export format and medications already stored in the fEMR database. The system will match medications by ID or exact full name, including strength and form.
 - a) (will require typing medication out - normalized on import)
- 2) Modifications require non-negative whole-number quantities, with the current quantity no greater than the initial quantity.
- 3) After upload, fEMR will display an editable preview showing additions, updates, unchanged rows, and conflicts.
 - a) Administrator must resolve unknown medications, duplicates, invalid quantities, and previously removed medications before confirming the import. Deleted CSV rows will not remove medications from the formulary, and invalid rows will never be silently skipped.
 - b) Final confirmation will clearly name the destination trip and summarize every proposed change.
- 4) Apply all changes in one database transaction, so either the full import succeeds or no inventory changes are saved.
 - a) The system will check the administrator's permission and trip assignment during upload, preview, and confirmation.

Questions

- What is the login information for the Test Nurse account?
- Can multiple accounts submit data to the same trip formulary simultaneously, i.e. will there be a race condition?
- Does anything in the kit require an internet connection at any point, and if so, what items need it?
- Who should be able to modify and import?